

CESE323 ERC1.0

Cloud Security Envisioning & Strategy

Student Table of Contents

Courseware Pack Front Matter | Instructor-led, Online, and Self-paced Delivery

Field	Value
Course code	CESE323
Version	ERC1.0
Audience	Students, delegates, hands-on lab participants
Delivery model	One-day instructor-led workshop with self-paced GitHub/LMS support
Primary technologies	Microsoft Defender for Cloud, Microsoft Defender XDR, Microsoft Entra ID, Microsoft Purview, Azure workload security, Microsoft Foundry / AI Services
Document purpose	Provide a structured student-facing table of contents for the Student Course Guide, Student Lab Guide, assessment pack, and online course site.

Student navigation principle

The course moves from security strategy and governance into Microsoft Defender for Cloud posture management, workload protection, AI security, live environment review, remediation, and evidence-based assessment.

Document Control

Item	Description
Document name	CESE323 Student Table of Contents
Version	ERC1.0
Prepared for	Skunkworks Academy courseware development
Student use	Guide navigation, lab sequence planning, assessment preparation, and self-paced learning orientation
Source basis	Workshop syllabus, Microsoft Learn collection, Microsoft security learning paths, pre-course lab readiness requirements

Courseware Structure at a Glance

Area	Student focus	Primary student artifact	Timing
Front Matter	Welcome, course overview, learning outcomes, prerequisites, student responsibilities, support model	Student Course Guide	Before class / first 15 minutes
Part 1	Strategic Initiation & Governance: goals, risk priorities, Microsoft security stack alignment, gap analysis	Student Course Guide + workbook activities	Instructor-led discussion
Part 2	Configuration Control & Workload Protection: Defender portal, CSPM, workload protection planning	Student Course Guide + Lab Guide	Guided demo + hands-on
Part 3	Practical Implementation & Remediation: configuration, policy assignment, AI-specific security, deployment blockers	Student Lab Guide + evidence pack	Hands-on lab block
Part 4	Hands-on Scenarios & Live Environment Review: incident scenarios, threat mapping, operational review	Scenario workbook + capstone activity	Scenario lab + review
Post-course	Cleanup, handover, operational next steps, feedback and learning path continuation	Cleanup checklist + feedback form	After class

Student Course Guide - Full Table of Contents

Page numbering note

The page column is intentionally marked as TBD because the final page numbers should be regenerated when the full Student Course Guide is assembled in Word, PDF, GitHub Pages, or LMS format.

No.	Section	Student-facing scope	Page
1	Welcome to CESE323	Course identity, purpose, delivery model, student expectations, and how to use the guide	TBD
1.1	Course overview	Cloud Security Envisioning & Strategy positioning and one-day workshop flow	TBD
1.2	Audience profile	Students, administrators,	TBD

		security engineers, solution architects, cloud engineers, and customer technical stakeholders	
1.3	Learning outcomes	Strategic alignment, unified visibility, threat reduction, workload hardening, and operational readiness	TBD
1.4	Courseware components	Student guide, lab guide, hints guide, assessment, evidence templates, feedback forms, and GitHub/LMS resources	TBD
1.5	How to use this guide	Icons, terminology, lab evidence markers, safety notes, and recommended reading sequence	TBD
2	Pre-Course Readiness	Required student preparation before attending the course	TBD
2.1	Minimum knowledge required	Cloud fundamentals, security fundamentals, Entra ID, Defender for Cloud, workload awareness, and AI security awareness	TBD
2.2	Student account readiness	Microsoft Learn, Azure portal, Defender portal, named user accounts, MFA, tenant selection, browser access	TBD
2.3	Azure subscription and lab access	Dedicated subscription, shared training subscription, customer live subscription, or demo-only model	TBD
2.4	Permissions and least privilege	Reader, Contributor, Security Admin, Owner boundaries and Global Administrator avoidance	TBD
2.5	Paid plan and budget awareness	Defender plan activation, cost controls, trial limitations, scope, budgets, and cleanup ownership	TBD
2.6	Local machine requirements	Laptop, stable internet, modern browser, PDF reader, screenshot tool, optional CLI tools	TBD
2.7	Student pre-course checklist	Final confirmation checklist to complete 5-7 days before class	TBD
3	Security, Compliance, and Identity Foundation	Baseline concepts required before strategy and hands-on activities	TBD
3.1	Shared responsibility model	Cloud responsibility boundaries for provider and customer	TBD
3.2	Defense in depth	Layered security controls across identity, network, platform, application, and data	TBD
3.3	Zero Trust model	Verify explicitly, use least privilege, and assume breach	TBD
3.4	Encryption and hashing	Data protection fundamentals for cloud services and workloads	TBD
3.5	Governance, risk, and compliance concepts	Security governance, regulatory posture, controls, and evidence	TBD
3.6	Identity concepts	Authentication, authorization,	TBD

		directory services, federation, and identity provider roles	
4	Microsoft Entra and Identity Security	Identity as the control plane for cloud and AI workloads	TBD
4.1	Microsoft Entra ID overview	Cloud identity and access management for apps, devices, users, and data	TBD
4.2	Identity types	Human identities, workload identities, external identities, hybrid identities, and agent identities	TBD
4.3	Authentication capabilities	Authentication methods, MFA, SSPR, password protection, and authentication strength	TBD
4.4	Access management fundamentals	Azure roles, managed identities, RBAC, Key Vault access, and scope management	TBD
4.5	Conditional Access basics	Security defaults, Conditional Access policies, session controls, and risk-based access concepts	TBD
5	Microsoft Security Solutions Overview	Microsoft security stack components used in the course	TBD
5.1	Microsoft Defender portal and XDR	Unified incident management, correlation, alert review, and automated response concepts	TBD
5.2	Microsoft Defender for Cloud	Cloud security posture management, workload protection, security policies, recommendations, and secure score	TBD
5.3	Microsoft Sentinel and SecOps context	SIEM/SOAR positioning, detection, response, and security workflow integration	TBD
5.4	Microsoft Purview security and compliance context	Information protection, DLP, Insider Risk, audit, eDiscovery, and data security posture	TBD
5.5	Microsoft Security Copilot context	Prompting, Copilot terminology, and role in security investigation workflows	TBD
6	Part 1 - Strategic Initiation & Governance	Workshop strategy block: goals, priorities, architecture value, and gap analysis	TBD
6.1	Identifying goals and priorities	Map business objectives, security outcomes, risk vectors, and stakeholder priorities	TBD
6.2	Solution architecture overview	Evaluate Microsoft security stack value proposition and market differentiators	TBD
6.3	Security gap analysis	Align cloud posture, compliance requirements, and business risk mitigation priorities	TBD
6.4	Student activity: security goal canvas	Capture top risks, target outcomes, and required security capabilities	TBD
7	Part 2 - Configuration Control & Workload Protection	Defender portal, CSPM, exposure management, and workload protection planning	TBD
7.1	Microsoft Defender portal	Unified XDR interface, incident	TBD

	deep dive	queues, alert correlation, investigation, and response	
7.2	CSPM overview	Secure Score, recommendations, asset inventory, workbooks, and regulatory compliance	TBD
7.3	Security Exposure Management	Attack path mapping, over-privileged identities, misconfigurations, and exposure reduction	TBD
7.4	Workload protection plan selection	Choose 2 of 4: Storage, Containers, App Service, or AI Services	TBD
7.5	Student decision matrix	Select workload tracks based on business risk, technical readiness, permissions, and cost control	TBD
8	Defender for Cloud: Security Posture Management	Hands-on posture discovery and risk prioritization	TBD
8.1	Connect hybrid and multicloud environments	Azure, on-premises, AWS, and GCP connector strategy and coverage validation	TBD
8.2	Identify security risks using CSPM	Risk-based recommendations, attack paths, choke points, and Cloud Security Explorer queries	TBD
8.3	Discover unprotected assets and vulnerabilities	External Attack Surface Management discovery and prioritization	TBD
8.4	Evaluate regulatory compliance	Control gaps, assigned standards, compliance dashboard, and reporting evidence	TBD
8.5	Enable and configure workload protection plans	CWPP plan catalog, environment settings, plan-specific components, and coverage workbook	TBD
9	Workload Protection Track A - Defender for Storage	Storage account protection, malware scanning awareness, suspicious access patterns, and data-layer risk	TBD
9.1	Storage protection overview	Protected workloads, benefits, and how Defender for Storage functions	TBD
9.2	Storage lab readiness	Azure subscription, Defender for Cloud, storage account, blob container, test files, permissions, and cost controls	TBD
9.3	Student lab: configure Storage workload protection	Enable plan, verify coverage, upload safe test files, review recommendations and alerts where available	TBD
10	Workload Protection Track B - Defender for Containers	AKS and Kubernetes workload protection concepts and configuration readiness	TBD
10.1	Container protection overview	AKS hardening, Kubernetes API access, registry scanning, runtime threat detection, and agent considerations	TBD
10.2	Container lab readiness	AKS cluster, supported version, network access, registry, permissions, and cost control	TBD

10.3	Student lab: review container posture	Assess cluster configuration, recommendations, coverage, and remediation options	TBD
11	Workload Protection Track C - Defender for App Service	Cloud-native application posture and runtime protection concepts	TBD
11.1	App Service protection overview	App runtime protection, recommendations, logs, and threat detection context	TBD
11.2	App Service lab readiness	Sample app, App Service plan, diagnostics, isolated resource group, and Defender plan selection	TBD
11.3	Student lab: assess App Service security posture	Review secure configuration, diagnostic settings, recommendations, and operational risks	TBD
12	Workload Protection Track D - Defender for AI Services	AI workload threat detection, posture, dashboard review, and prompt/data leakage risk	TBD
12.1	AI workload risk overview	Gen-AI threat model, data leakage, prompt injection, model behavior risk, and runtime detection	TBD
12.2	Enable Defender for AI Services plan	Plan activation, components, Data and AI security dashboard, and alert investigation	TBD
12.3	Microsoft Foundry protection	AI guardrails, Defender for Cloud AI Workloads plan, CWP/CSPM insights, and Defender XDR prompt evidence	TBD
12.4	AI identity infrastructure	Management plane/data plane access, workload identities, RBAC, Conditional Access, and identity misconfigurations	TBD
12.5	Student lab: review AI security posture	Enable/observe AI plan, review dashboard, validate safe prompt/data handling, and identify remediation actions	TBD
13	Part 3 - Practical Implementation & Remediation	Configuration, policy assignment, deployment friction, and remediation planning	TBD
13.1	Deep dive into selected modules	Practical configuration and policy assignment for the selected Defender components	TBD
13.2	AI-specific security	Shared responsibility in AI, data leakage monitoring, prompt injection considerations, and safe use boundaries	TBD
13.3	Technical remediation	Deployment blockers, permissions gaps, connector issues, unsupported services, and remediation tracking	TBD
13.4	Student activity: remediation backlog	Prioritize findings by risk, impact, owner, effort, and deadline	TBD
14	Part 4 - Hands-on Scenarios & Live Environment Review	Scenario demonstrations, threat mapping, and	TBD

		operational review	
14.1	Key scenario demonstrations	Simulated security incidents and product capability demonstrations	TBD
14.2	Threat mapping	Map discovered vulnerabilities to Microsoft-recommended mitigations and architectural controls	TBD
14.3	Interactive operational review	Review environment settings, Defender coverage, recommendations, and tailored operational examples	TBD
14.4	Live environment safety controls	Written approval, named scope, no production disruption, change window, rollback, audit trail, and data handling rules	TBD
15	Student Assessment and Evidence Requirements	Knowledge checks, lab evidence, scenario analysis, and capstone review	TBD
15.1	Knowledge assessment overview	Multiple-choice, terminology, scenario, and architecture judgment questions	TBD
15.2	Lab evidence requirements	Screenshots, configuration notes, observations, recommendations, and remediation decisions	TBD
15.3	Capstone scenario	Security posture review, workload protection selection, and executive remediation recommendation	TBD
15.4	Reflection and action plan	Next steps, operational owners, technical remediation backlog, and post-course learning path	TBD
16	Post-Course Cleanup and Handover	Cost control, access cleanup, and operational transition	TBD
16.1	Lab cleanup checklist	Disable paid plans where required, delete lab resources, remove temporary access, and review spend	TBD
16.2	Operational handover checklist	Export key findings, document scope, assign remediation owners, and define review cadence	TBD
16.3	Course feedback form	Student feedback, course quality, lab readiness, pacing, and improvement suggestions	TBD
17	Appendices	Reference material and reusable student resources	TBD
17.1	Glossary and acronyms	CSPM, CWPP, XDR, SIEM, SOAR, EASM, RBAC, MFA, DLP, GRC, MCRA, MCSB, CAF, WAF	TBD
17.2	Portal and resource links	Azure portal, Defender portal, Entra admin center, Microsoft Learn, GitHub/course site, Teams/Loop/LMS	TBD
17.3	Student worksheet templates	Risk canvas, workload selection matrix, remediation backlog, lab evidence log, and cleanup checklist	TBD
17.4	Recommended self-paced learning continuation	SC-900 foundations, Defender for Cloud learning paths, AI	TBD

		security modules, SC-100 architecture extensions	
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Student Lab Guide - Table of Contents

Lab	Title	Student task focus	Expected evidence
Lab 0	Environment Access Verification	Confirm Microsoft Learn, Azure portal, Defender portal, tenant, MFA, permissions, and assigned subscription/resource group.	Access proof screenshot + readiness confirmation
Lab 1	Security Goals and Risk Priorities	Map business goals to cloud security risks, compliance requirements, and Defender capability priorities.	Completed risk canvas
Lab 2	Defender for Cloud Onboarding and Secure Score	Verify Defender for Cloud access, Environment settings, Secure Score, recommendations, inventory, and workbooks.	Secure Score screenshot + recommendation notes
Lab 3	CSPM Recommendations, Attack Paths, and Regulatory Compliance	Identify misconfigurations, over-privileged identities, attack paths, and compliance gaps.	Prioritized findings table
Lab 4	Workload Protection Planning	Select 2 workload protection modules based on risk, readiness, permissions, budget, and technical fit.	Workload selection decision matrix
Lab 5A	Defender for Storage	Configure or review Storage protection and related recommendations using safe test data only.	Coverage/configuration evidence
Lab 5B	Defender for Containers	Review AKS/container posture, coverage requirements, recommendations, and remediation options.	Cluster posture review notes
Lab 5C	Defender for App Service	Assess App Service security posture, logging, diagnostics, and Defender recommendations.	App workload posture evidence
Lab 5D	Defender for AI Services / Microsoft Foundry	Review AI workload posture, AI service protection, prompt/data handling controls, and identity risk.	AI security dashboard or simulated review evidence
Lab 6	Scenario-based Threat Mapping	Map selected vulnerabilities or alerts to Microsoft-recommended mitigations and Zero Trust controls.	Threat-to-mitigation map
Lab 7	Live Environment Review / Demo Review	Walk through Defender settings, coverage, security recommendations, and operational examples in a controlled scope.	Operational review notes
Lab 8	Remediation Backlog and Cleanup	Create a remediation backlog and complete lab cleanup, access removal, resource deletion, and cost review.	Remediation backlog + cleanup checklist

Student Assessment Pack - Table of Contents

Item	Assessment component	Scope	Evaluation method
A1	Pre-course readiness confirmation	Completion check before class begins	Pass / remediation required
A2	Knowledge Check 1: Security, compliance, and identity foundations	Shared responsibility, Zero Trust, defense-in-depth, identity, encryption, GRC	Auto-graded or instructor-reviewed
A3	Knowledge Check 2: Defender for Cloud and CSPM	Secure Score, recommendations, asset inventory, compliance, attack paths, exposure management	Auto-graded or instructor-reviewed
A4	Knowledge Check 3: Workload protection and AI security	Storage, Containers, App Service, AI Services, Foundry, workload identities, safe data handling	Auto-graded or instructor-reviewed
A5	Lab evidence submission	Screenshots and notes proving participation and technical review	Rubric-based evidence review
A6	Capstone scenario	Recommend security posture improvements and workload protection selections for a sample customer environment	Scenario rubric
A7	Course feedback form	Student evaluation of pace, relevance, lab readiness, materials, and delivery quality	Feedback summary

Self-Paced Web / GitHub Course Site - Proposed Navigation

Path	Page title	Purpose
/docs/course-overview.mdx	Course Overview	One-page student overview with outcomes, audience, prerequisites, agenda, and learning path links
/docs/pre-course/readiness.mdx	Pre-Course Readiness	Student access checklist, Azure subscription requirements, permissions, MFA, and lab safety
/docs/modules/security-foundations.mdx	Security Foundations	Shared responsibility, Zero Trust, defense in depth, identity concepts, governance, risk, and compliance
/docs/modules/entra-identity.mdx	Microsoft Entra and Identity Security	Identity types, authentication, MFA, RBAC, Conditional Access, workload identities, and AI identity considerations
/docs/modules/defender-cloud-cspm.mdx	Defender for Cloud and CSPM	Secure Score, recommendations, asset inventory, regulatory compliance, attack paths, and exposure management
/docs/modules/workload-protection.mdx	Workload Protection Tracks	Storage, Containers, App Service, AI Services, Microsoft Foundry, and selected lab tracks
/docs/labs/index.mdx	Student Labs	Lab index, evidence requirements, screenshots, safety notes, and cleanup instructions
/docs/assessments/knowledge-check.mdx	Knowledge Assessment	Student knowledge checks, scenario questions, and self-assessment guidance
/docs/assessments/capstone-scenario.mdx	Capstone Scenario	Practical security posture review and remediation recommendation exercise
/docs/resources/index.mdx	Resources and Appendices	Glossary, portal links, Microsoft Learn links, templates, and post-course learning path

Recommended Student Workbook Templates

Template	Name	Suggested fields
Template 1	Security Goal Canvas	Business goal, risk statement, impacted assets, existing controls, desired outcome, owner
Template 2	Cloud Security Posture Review Log	Secure Score observation, recommendation, severity, affected resource, owner, remediation action
Template 3	Workload Protection Selection Matrix	Workload type, business value, threat exposure, readiness, permissions, estimated cost, selected/not selected
Template 4	Lab Evidence Log	Lab number, screenshot reference, observation, result, issue found, action taken
Template 5	Remediation Backlog	Finding, risk rating, business impact, effort, priority, owner, target date, status
Template 6	Cleanup Checklist	Resource cleanup, paid plan review, role assignment removal, budget check, final sign-off

Optional Advanced Extension Topics

Extension	Student scope
SC-100 Architecture Extension	Zero Trust architecture, CAF/WAF alignment, MCRA, MCSB, infrastructure, application/data security, SecOps, identity, and compliance design
Microsoft Purview Extension	Data classification, sensitivity labels, DLP, insider risk, audit, eDiscovery, adaptive protection, and data security posture management
AI Security Extension	Microsoft Foundry security, AI guardrails, Defender for AI workloads, AI identity architecture, managed identities, RBAC, and Conditional Access
SecOps Extension	Microsoft Defender XDR, Sentinel, SIEM/SOAR, centralized logging, detection engineering, incident workflow, and response automation
Hybrid and Multicloud Extension	Azure Arc, AWS/GCP connectors, multicloud posture management, EASM, and hybrid workload protection

Final Assembly Notes for the Student Course Guide

- Keep this TOC as the master structure for the Student Course Guide, Student Lab Guide, and GitHub/LMS navigation.
- Generate final page numbers only after the full course guide is assembled, reviewed, and exported to PDF.
- For the one-day workshop, treat SC-900 and SC-100 content as targeted support material, not full-course replacement content.
- Only run live environment activities where written scope, permissions, cost approval, and rollback procedures are in place.
- Use demo or sandbox environments for AI, container, and paid workload protection labs unless the customer has explicitly approved live configuration changes.

Source Alignment Summary

Source group	How it informs this TOC
Workshop syllabus	Used to structure objectives, four-part agenda, workload protection selection, practical remediation, and live environment review.
Microsoft Learn collection	Used to align student learning navigation with Microsoft security, Defender for Cloud, Entra, Purview, AI security, and architecture learning paths.
Pre-course lab requirements	Used to include readiness, permissions, Azure subscription, Defender plan, paid plan, lab safety, evidence, and cleanup content.